SMARTPICK AI

Test Plan

Functional, data-quality, ranking, and user-interface verification

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Status: Working companion document

1. Test objectives

- Verify that data validation, budget filtering, preference weighting, ranking, explanations, and error states work as specified.

2. Test cases

- TC-01 Valid dataset: required columns are present and valid records load successfully.
- TC-02 Missing required column: app reports the exact missing column rather than crashing.
- TC-03 Budget filtering: every returned phone is within the user’s selected budget.
- TC-04 No eligible phones: app shows a no-results message.
- TC-05 Weights: negative weights and all-zero weights are rejected.
- TC-06 Ranking: returned results are ordered by descending score with a deterministic tie-break.
- TC-07 Missing optional field: dependent filter is hidden or disabled; values are not fabricated.
- TC-08 Empty dataset: app shows a controlled message.
- TC-09 Recommendation explanation: every claim maps to an available field.
- TC-10 Reproducibility: same input and dataset return the same ranked output.
- TC-11 Evaluation: Precision@K/Recall@K/NDCG@K are computed only with relevant labels.
- TC-12 UI smoke test: app launches and a valid profile produces visible results.

3. Exit criteria

- All must-have test cases pass or have documented limitations.
- No displayed result violates hard budget constraints.
- Known issues are listed in README and presentation.